ISABELLA RIVERA

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Education

Honours Bachelor of Science, University of Toronto — St. George Campus *Expected June 2027*

Specialist in Astronomy and Physics; Minor in Mathematics

Research Experience

Radio Search for Off-Axis GRB Jets — Undergraduate Thesis *September 2026 – Present*

Dunlap Institute for Astronomy & Astrophysics · Supervised by Prof. Maria Drout and Dr. Genevieve Schroeder

- Search VLASS data for late-time radio emission from off-axis relativistic jets associated with Type Ic supernovae.
- Model radio light curves to constrain jet energetics and viewing angles, and assess which supernovae may have hosted off-axis GRBs.

FRB Multi-wavelength Counterpart Searches — SURP *May 2026 – Present*

Dunlap Institute for Astronomy & Astrophysics · Supervised by Prof. Maria Drout, Prof. Juan Mena-Parra, and Mattias Lazda

- Searching for persistent radio sources (PRSs) associated with fast radio bursts (FRBs) within the CHIME/FRB Collaboration.
- Use archival radio and optical surveys to identify candidate counterparts and characterize the local environments of FRBs.

CHIME/FRB Far-Sidelobe Localization — AST431 (A+) *Sept 2025 – June 2026*

David A. Dunlap Department of Astronomy & Astrophysics · Supervised by Prof. Juan Mena-Parra

- Developed and tested localization techniques for FRBs detected in CHIME's far sidelobes by characterizing instrumental systematics and evaluating geometric corrections.
- Applied improved localizations to support probabilistic host-galaxy association and follow-up.

Tracing Star Formation Histories in Vela — AST430 (A+) *Jan 2026 – May 2026*

David A. Dunlap Department of Astronomy & Astrophysics · Supervised by Dr. Ronan Kerr and Prof. Joshua Speagle (SPYGLASS program)

- Investigated star-forming regions in the Vela Complex, studying their morphology and evolutionary histories.
- Used Gaia astrometry, stellar age fitting, and orbital traceback to identify possible shared formation histories among young stellar populations.

Radio Astronomy Research Assistant *May 2025 – Sept 2025*

David A. Dunlap Department of Astronomy & Astrophysics · Supervised by Prof. Juan Mena-Parra

- Developed and tested clock synchronization methods for the CHIME Outriggers using noise modelling, interpolation, and timing-error analysis.
- Explored a new localization approach for mitigating timing uncertainties between telescope stations.

Publications

Gusinskaia, N., et al. incl. **Rivera, I.** (5th author) (2026). Improved Localization of CHIME/FRB Far-Sidelobe Fast Radio Bursts. In preparation.

Dong, V., **Rivera, I.***, et al. (The CHIME/FRB Collaboration) (2026). A Comprehensive Search for Multiwavelength Counterparts to the Second CHIME/FRB Baseband Catalog. In preparation.

*Co-second author

Awards

Professor Mercedes T. Richards Award for Excellence in Summer Undergraduate Research — $2,000 *September 2026*

SURP, University of Toronto

- Awarded to one SURP student demonstrating outstanding achievement and growth in research, evaluated across technical/research skills, independence of thought, and science communication.

- Selected from 15 nominations and 30+ students.

The Roy Alvin Hope Scholarship — $1,000 *August 2026*
Victoria University, University of Toronto

- Awarded to students who achieved overall A standing in third year.

Student's Poster Pick Award *August 2026*
SURP Poster Fair, University of Toronto

- Awarded for a poster presenting research on persistent radio sources associated with FRBs.

Dragonfly Summer Research Fellowship — $5,500 *May 2026*
David A. Dunlap Department of Astronomy and Astrophysics

- Funded by the Dragonfly Focused Research Organization to support undergraduate research in astrophysics, particularly extragalactic astronomy.

Dunlap Admin's Bestie Award *January 2026*
David A. Dunlap Department of Astronomy and Astrophysics

- Awarded to consistent, dependable outreach volunteers at the Dunlap Institute.
- First undergraduate recipient in the department's history.

John L. Horwood Memorial Scholarship — $1,000 *August 2025*
David A. Dunlap Department of Astronomy and Astrophysics

- Awarded to a student with overall A standing in second year, with preference for astronomy coursework.

Samuel Beatty Travel Award — $2,000 *January 2025*
University of Toronto Physics Department

Admissions Scholarship — $1,000 *September 2023*
University of Toronto

Presentations

Talks

AST201: Stars and Galaxies — Guest Lecturer *August 2026*
University of Toronto

Physics Union Undergraduate Research Conference — Talk *March 2026*
University of Toronto

"One Neat Thing" — Talk *December 2025*
University of Toronto

Posters

Dynamic Radio Sky (DRS) — Flash Talk & Poster *August 2026*
McGill University

SURP Poster Fair — Poster (Student's Poster Pick Award) *August 2026*
University of Toronto

Canadian Astronomical Society (CASCA) Annual Meeting — Poster *June 2026*
Université de Montréal

Research Opportunity Program (ROP) Poster Fair — Poster *April 2025*
University of Toronto

Outreach & Leadership

President, Astronomy & Space Exploration Association (ASX) *April 2025 – Present*
University of Toronto

- Symposium Director, ASX Symposium (2025–2026)
- Events Director, ASX (2024–2025)

Volunteer, AstroTours & Astro on TAP *Sept 2024 – Present*
University of Toronto – Department of Astronomy and Astrophysics

External Affairs Director, Women in Science and Engineering (WISE) *April 2024 – April 2025*

University of Toronto

Volunteer, Girls in STEM Workshop Event *May 2024*

University of Toronto – Department of Astronomy and Astrophysics

Events Assistant, Work Study Program *April 2024 – August 2024*

Dunlap Institute for Astronomy and Astrophysics

Host *April 2024 – September 2024*

Ontario Science Centre

Skills

- **Technical:** Python, NumPy, SciPy, Pandas, Git, LaTeX, command-line workflows. Completed Computational Astrophysics (CTA200, University of Toronto) with a final grade of an A+.
- **Research Areas:** Radio astronomy, time-domain and transient astrophysics (fast radio bursts, gamma-ray bursts), multi-wavelength astronomy, and stellar evolution.
- **Communication & Leadership:** Scientific literature synthesis, research presentations, scientific communication, and leadership experience.
- **Languages:** English; beginner Spanish.